Math 110 Hard Mock Midterm 2

Solutions

Question 1

Work in the basis $1,\ x-1,\ (x-1)^2$, which is adapted to the point 1. Write $p = a+b(x-1)+c(x-1)^2$. Then

$$\varphi_1(p) = a, \qquad \varphi_2(p) = b, \qquad \varphi_3(p) = \int_{-1}^{1} (a + bt + ct^2)\, dt = 2a + \tfrac{2}{3}c$$

(substitute $t = x - 1$; the odd term integrates to 0).

(a)

We need $\varphi_i(p_j) = 1$ if $i = j$ and 0 otherwise.

- p_1: $a = 1$, $b = 0$, $2 + \frac{2}{3}c = 0$, so $c = -3$: $\quad p_1 = 1 - 3(x-1)^2$.
- p_2: $a = 0$, $b = 1$, $\frac{2}{3}c = 0$: $\quad p_2 = x - 1$.
- p_3: $a = 0$, $b = 0$, $\frac{2}{3}c = 1$: $\quad p_3 = \frac{3}{2}(x-1)^2$.

If $c_1p_1 + c_2p_2 + c_3p_3 = 0$, applying φ_i gives $c_i = 0$. So p_1, p_2, p_3 is independent, hence a basis of the 3-dimensional V. Since $\varphi_i(p_j)$ is 1 for $i = j$ and 0 otherwise, $\varphi_1, \varphi_2, \varphi_3$ is the dual basis of p_1, p_2, p_3, so it is a basis of V'.

(b)

$U = \operatorname{null}\varphi_1 \cap \operatorname{null}\varphi_3$. For $p = \sum c_jp_j$ we have $\varphi_i(p) = c_i$, so $p \in U$ iff $c_1 = c_3 = 0$:

$$U = \operatorname{span}(p_2) = \operatorname{span}(x - 1).$$

U^0 has dimension $3 - 1 = 2$ and contains φ_1 and φ_3 (both kill U), which are independent. So

$$U^0 = \operatorname{span}(\varphi_1, \varphi_3).$$

Trap. U^0 is a subspace of V'. Its basis is a list of functionals, not polynomials.

(c)

Use $\psi = \sum_j \psi(p_j)\varphi_j$ for $\psi \in V'$.

- $D'\varphi_1(p) = p'(1) = \varphi_2(p)$, so $D'\varphi_1 = \varphi_2$.

- $D'\varphi_2(p) = p''(1)$. Values on p_1, p_2, p_3: $-6,\ 0,\ 3$. So $D'\varphi_2 = -6\varphi_1 + 3\varphi_3$.
- $D'\varphi_3(p) = \int_0^2 p' = p(2) - p(0)$. Values: p_1: $(-2) - (-2) = 0$; p_2: $1 - (-1) = 2$; p_3: $\frac{3}{2} - \frac{3}{2} = 0$. So $D'\varphi_3 = 2\varphi_2$.

Now $D'(a\varphi_1 + b\varphi_2 + c\varphi_3) = -6b\,\varphi_1 + (a + 2c)\varphi_2 + 3b\,\varphi_3$, which is 0 iff $b = 0$ and $a = -2c$. So

$$\operatorname{null} D' = \operatorname{span}(\varphi_3 - 2\varphi_1).$$

Check: $\operatorname{range} D = \mathcal{P}_1(\mathbb{R})$, so $(\operatorname{range} D)^0$ has dimension $3 - 2 = 1$, and $\varphi_3 - 2\varphi_1$ kills 1 $(2 - 2 = 0)$ and x $(2 - 2 = 0)$. So $(\operatorname{range} D)^0 = \operatorname{span}(\varphi_3 - 2\varphi_1) = \operatorname{null} D'$.

Trap. $D'\varphi$ is $\varphi \circ D$: differentiate first, then apply φ.

Question 2

(a)

Let $v_1, \ldots, v_n$ be an eigenbasis and $u_1, \ldots, u_m$ a basis of U. Go through $v_1, \ldots, v_n$ in order and add v_i to the list $u_1, \ldots, u_m$ whenever it is not in the span of the list so far. The final list spans V (every v_i is in its span) and is independent, so it is a basis $u_1, \ldots, u_m, v_{i_1}, \ldots, v_{i_k}$. Let $W = \operatorname{span}(v_{i_1}, \ldots, v_{i_k})$. Then $V = U \oplus W$, and W is invariant because it is spanned by eigenvectors.

(b)

U is invariant (each eigenvector is mapped to a multiple of itself). By (C), $V = U \oplus W$ with W invariant. If $W \neq \{0\}$, then $T|_W$ is an operator on a nonzero complex space, so it has an eigenvector $w \in W$. Then w is an eigenvector of T, so $w \in U \cap W = \{0\}$: contradiction. So $W = \{0\}$ and $U = V$. A spanning list of eigenvectors contains a basis of eigenvectors, so T is diagonalizable.

Trap. The complex hypothesis is used exactly once: to get an eigenvector of $T|_W$.

(c)

Rotation $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on $\mathbb{R}^2$. An invariant line would be spanned by a real eigenvector, and R has none. So the only invariant subspaces are $\{0\}$ and $\mathbb{R}^2$, which are complements of each other: R has property (C). But R is not diagonalizable over $\mathbb{R}$.

(d)

$U = \operatorname{span}(e_1) = \operatorname{null} N$. A complement W would be a line; an invariant line is spanned by an eigenvector, and every eigenvector of N lies in $\operatorname{null} N = U$. So no invariant W has $U \cap W = \{0\}$.

Question 3

(a)

$D(a + bx) = b \in U$ and $E(a + bx) = a + b + bx \in U$, so U is invariant under both. Since $1, x$ is a basis of U and $1, x, x^2, x^3$ a basis of V, the list $x^2 + U,\ x^3 + U$ is a basis of V/U.

- $(D/U)(x^2+U) = 2x+U = 0+U$ and $(D/U)(x^3+U) = 3x^2+U$.
- $(E/U)(x^2+U) = x^2+2x+1+U = x^2+U$ and $(E/U)(x^3+U) = x^3+3x^2+3x+1+U = x^3+3x^2+U$.

$$\mathcal{M}(D/U) = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}, \qquad \mathcal{M}(E/U) = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}.$$

Trap. In V/U every term of degree at most 1 is 0: $2x+U = 0+U$ because $2x \in U$.

(b)

$\mathcal{M}(D/U)^2 = 0$ and $\mathcal{M}(D/U) \neq 0$, so $m_{D/U} = x^2$. $\big(\mathcal{M}(E/U) - I\big)^2 = 0$ and $\mathcal{M}(E/U) \neq I$, so $m_{E/U} = (x-1)^2$. Both have a repeated root, so neither is diagonalizable (Prop. 68).

(c)

Both sides are linear, so check them on x^k ($k \leq 3$). Since $D^j x^k = \frac{k!}{(k-j)!} x^{k-j}$ for $j \leq k$ and 0 for $j > k$,

$$\sum_{j=0}^{3} \frac{1}{j!} D^j x^k = \sum_{j=0}^{k} \binom{k}{j} x^{k-j} = (x+1)^k = Ex^k.$$

So $E - I = D + \frac{1}{2}D^2 + \frac{1}{6}D^3 = DQ$ with $Q = I + \frac{1}{2}D + \frac{1}{6}D^2$. Q commutes with D, and in the basis $1, x, x^2, x^3$ it is upper-triangular with 1's on the diagonal, so it is invertible (Prop. 60). Hence $(E-I)^k = D^k Q^k$. Since $D^3 x^3 = 6 \neq 0$ and Q^3 is invertible, $(E-I)^3 \neq 0$; and $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$, so $(E-I)^4 = 0$. Therefore m_E divides $(x-1)^4$ but is not $(x-1)^j$ for $j \leq 3$: $m_E = (x-1)^4$.

Trap. E has only the eigenvalue 1 but is not I, so it is not diagonalizable; m_E has degree $4 = \dim V$.

Question 4

(a)

$(ST)' = T'S'$ gives $(T^k)' = (T')^k$, the dual of I_V is $I_{V'}$, and $S \mapsto S'$ is linear, so $p(T)' = p(T')$. If $S' = 0$ but $Sv \neq 0$ for some v, the separating functional gives φ with $\varphi(Sv) = 1$, so $(S'\varphi)(v) = 1 \neq 0$: contradiction. So $S' = 0$ iff $S = 0$. Hence $p(T') = 0$ iff $p(T)' = 0$ iff $p(T) = 0$: exactly the same polynomials send T and T' to 0, so $m_{T'} = m_T$.

(b)

For $\varphi \in U^0$ and $u \in U$: $(T'\varphi)(u) = \varphi(Tu) = 0$ since $Tu \in U$. So $T'\varphi \in U^0$. The restriction satisfies $m_T(T'|_{U^0}) = m_{T'}(T')|_{U^0} = 0$ by (a), so its minimal polynomial divides m_T (Prop. 65).

(c)

2 and 1 are roots of m_T, hence eigenvalues (Prop. 66), so $U = E(2,T) \neq \{0\}$ and $E(1,T) \neq \{0\}$. If $\dim U \geq 2$, then $\dim E(1,T) + \dim E(2,T) \geq 3 = \dim V$ and T would be diagonalizable (Prop. 62), but m_T has a repeated root (Prop. 68). So $\dim U = 1$.

U is invariant and $T|_U = 2I$, so $m_{T|_U} = x-2$. Since m_T divides $m_{T|_U} m_{T/U}$ (prep packet T5(1); notes, proof of Prop. 67), m_T divides $(x-2)\, m_{T/U}$, so $(x-1)^2$ divides $m_{T/U}$. Also $\deg m_{T/U} \leq \dim V/U = 2$. So $m_{T/U} = (x-1)^2$.

Trap. The key step is "m_T divides $m_{T|_U} m_{T/U}$". Without it you only know $m_{T/U}$ divides m_T, which allows $x-1$.